Classical Field Theory — Exercise Sheet 16

Energy, plateaux and local blow-up

Solutions

Use the general one-dimensional local theorem proved on Sheet 15: for $V \in C^3$, $V'(0) = 0$, data in $H^1(\mathbb{R}) \times L^2(\mathbb{R})$ have a unique weak energy solution, and a finite maximal endpoint forces its full energy-space norm to tend to infinity. Its reference linear mass is one. The conservation argument below must be made at that weak regularity.

Exercise 16.1 (A nonnegative potential beyond the quartic). Let $V \in C^3(\mathbb{R})$ satisfy $V \geq 0$ and $V(0) = V'(0) = 0$. For the energy solution of $u_{tt} - u_{xx} + V'(u) = 0$ with data (u_0, g_0), set

$$E(t) = \int_{\mathbb{R}} \left[\frac{1}{2} u_t^2 + \frac{1}{2} u_x^2 + V(u) \right] \mathrm{d}x.$$

(a) Prove finiteness, continuity and conservation of E by smoothing the actual solution in space. Justify the potential chain rule and the endpoint limits. Do not assume a density theorem for nonlinear solutions.

(b) Prove global existence and give a finite-time bound on the full $H^1 \times L^2$ norm. For $V(s) = s^2/2 + s^6/6$, obtain a stronger bound uniform in time. Identify which part of the proof needs $V \geq 0$.

Solution. For the first part, fix a compact time interval K inside the lifespan. The one-dimensional embedding bounds $|u|$ by some M on $K \times \mathbb{R}$. Since $V(0) = V'(0) = 0$, boundedness of V'' on $[-M, M]$ gives

$$|V(a)| \leq C_M a^2, \quad |V'(a)| \leq C_M |a|, \quad |V(a) - V(b)| \leq C_M (|a| + |b|)|a - b|.$$

It follows that $V(u) \in C(K; L^1)$ and $V'(u) \in C(K; L^2)$. Thus the initial integral and the time-dependent energies are finite and continuous.

Choose a nonnegative compact smooth unit-mass mollifier and denote spatial convolution by a subscript ε. Compactness of the time images and the approximate-identity theorem give

$$u_\varepsilon \to u \text{ in } C(K; H^1), \qquad p_\varepsilon \to u_t \text{ in } C(K; L^2), \qquad |u_\varepsilon| \leq M.$$

The smoothed equation is

$$\partial_t p_\varepsilon = \partial_x^2 u_\varepsilon - (V'(u))_\varepsilon, \qquad (u_\varepsilon)_t = p_\varepsilon, \qquad R_\varepsilon = V'(u_\varepsilon) - (V'(u))_\varepsilon.$$

For fixed ε, the right side of the first equation is continuous in L^2. The norm-primitive argument makes p_ε continuously norm differentiable there. Also $\partial_t \partial_x u_\varepsilon = \partial_x p_\varepsilon \in C(K; L^2)$.

To justify the potential derivative, Taylor's theorem on the bounded range gives

$$\left| \int [V(u_\varepsilon(t+h)) - V(u_\varepsilon(t)) - V'(u_\varepsilon(t))(u_\varepsilon(t+h) - u_\varepsilon(t))] \mathrm{d}x \right| \leq C_M \|u_\varepsilon(t+h) - u_\varepsilon(t)\|_2^2.$$

The right side is $O(h^2)$ because u_ε is C^1 in L^2. Divide by h and use $V'(u_\varepsilon(t)) \in L^2$ to obtain $\frac{\mathrm{d}}{\mathrm{d}t} \int V(u_\varepsilon) = \int V'(u_\varepsilon) p_\varepsilon$. The derivative is continuous by the same Lipschitz bound on V'. The kinetic and spatial squared norms are also differentiable. Integration by parts with large spatial cutoffs leaves an error at most $C\|p_\varepsilon\|_2 \|\partial_x u_\varepsilon\|_2 / R$, tending to zero. Consequently

$$\frac{\mathrm{d}}{\mathrm{d}t} E[u_\varepsilon, p_\varepsilon] = \int p_\varepsilon R_\varepsilon \mathrm{d}x.$$

The residual tends to zero uniformly in $C(K;L^2)$: split it into $V'(u_\varepsilon)-V'(u)$ and $V'(u)-(V'(u))_\varepsilon$. The first is bounded by $C_M\|u_\varepsilon - u\|_2$ and the second converges by the compact-image approximate-identity argument. The velocities are uniformly bounded. Integrate between any two times in K and pass to the limit. The potential difference estimate above gives uniform convergence of the endpoint integrals, while the two quadratic terms converge by the norm limits. Hence $E(t) = E(s)$, including $s = 0$ whenever it is one of the endpoints. No approximation by exact smooth nonlinear solutions was used.

For the second part, positivity and conservation give $\|u_t\|_2, \|u_x\|_2 \le \sqrt{2E_0}$. The L^2 primitive gives

$$\|u(t)\|_2 \le \|u_0\|_2 + |t|\sqrt{2E_0}, \qquad \|(u(t),u_t(t))\|_{\mathrm{lin}}^2 \le \big(\|u_0\|_2 + |t|\sqrt{2E_0}\big)^2 + 4E_0.$$

Every finite-time endpoint is excluded by the continuation alternative. For $V(s) = s^2/2+s^6/6$, the energy is at least half the entire squared linear norm, so that squared norm is bounded by $2E_0$ for all time. The conservation proof used bounded derivatives on the solution range and the two vanishing conditions at zero. Nonnegativity is needed to extract the bounds that imply continuation, not to establish conservation itself.

Exercise 16.2 (Sine-Gordon energy does not bound every phase norm). Take $V(u) = 1 - \cos u$ and background zero.

(a) Use Problem 1 to prove energy conservation and global energy-class evolution. Verify explicitly the initial potential integrability and continuity needed for that argument.

(b) Choose a fixed nondecreasing $\eta \in C^\infty(\mathbb{R})$, zero for $s \le 0$ and one for $s \ge 1$. For $R \ge 1$ take initial displacement

$$u_R(x) = 2\pi\eta(x + R + 1)\eta(R + 1 - x)$$

and zero velocity. Compute its energy in terms of two integrals on $[0, 1]$ and show that it is independent of R, while $\|u_R\|_2 \to \infty$. Does this contradict global existence or the finite-time bound of Problem 1?

Solution. For the first part, the potential is nonnegative and smooth with $V(0) = V'(0) = 0$. Its integral is finite since $0 \le 1 - \cos u \le u^2/2$. Moreover, the fundamental theorem of calculus and $|\sin s| \le |s|$ give

$$|V(a) - V(b)| \le (|a| + |b|)|a - b|.$$

Cauchy–Schwarz proves continuity in L^1 along an L^2-continuous energy path. All hypotheses of Problem 1 apply. Its conservation proof and continuation bounds give global evolution, in agreement with Sheet 15's independent weighted-contraction construction.

For the second part, $u_R = 2\pi$ on $[-R, R]$ and is zero outside $[-R - 1, R + 1]$. On the left ramp it is $2\pi\eta(s)$ with $s = x + R + 1 \in [0, 1]$; the right ramp has the reversed coordinate and the same energy. The plateau and exterior contribute no gradient or potential. Thus

$$E[u_R, 0] = 4\pi^2 \int_0^1 \eta'(s)^2 \mathrm{d}s + 2\int_0^1 [1 - \cos(2\pi\eta(s))]\mathrm{d}s.$$

This is a fixed finite positive number. However, integrating the squared plateau gives

$$\|u_R\|_2^2 \ge \int_{-R}^{R} 4\pi^2 \mathrm{d}x = 8\pi^2 R \longrightarrow \infty.$$

Each datum is compact and smooth and has its own finite displacement norm. Problem 1's bound retains that initial norm, so it does not claim a common bound depending only on energy. The family contradicts no continuation result.

Exercise 16.3 (Focusing blow-up from compact data on the line). Change the potential to $V(u) = -u^4/4$ on $\mathbb{R}$, so $u_{tt} - u_{xx} = u^3$. Fix $0 < T < R$ and choose a real $\chi \in C_c^\infty(\mathbb{R})$ with $0 \le \chi \le 1$ and $\chi = 1$ on $[-R, R]$. Take

$$u_0 = \frac{\sqrt{2}}{T}\chi, \qquad g_0 = \frac{\sqrt{2}}{T^2}\chi.$$

For this conditional exercise, assume additionally that the unique energy solution for the displayed data has a C^2 representative throughout its maximal energy-class lifespan $[0, T_*)$. This is a regularity hypothesis, and assumes no global existence.

(a) Compute its finite initial focusing energy. Compare on the cone $|x| \le R - t$ with $q(t) = \sqrt{2}/(T - t)$ for every common time $0 \le t < \min\{T, T_*\}$. Prove the needed local uniqueness rather than assuming a global finite-energy theorem for q.

(b) Deduce $T_* \le T$. State carefully what was proved if the maximal solution terminates earlier than T. Why do these data improve on simply placing the spatially constant ODE on all of $\mathbb{R}$?

Solution. For the first part, direct substitution of the data gives

$$E_0 = \int \left[\frac{1}{2}g_0^2 + \frac{1}{2}(u_0')^2 - \frac{1}{4}u_0^4\right] \mathrm{d}x = T^{-2}\int \chi'^2 \mathrm{d}x + T^{-4}\int (\chi^2 - \chi^4)\mathrm{d}x.$$

Both integrals are finite and nonnegative. The cutoff is nonconstant, so the derivative integral is positive. Thus the data have positive finite energy and belong to $H^1 \times L^2$. The local theorem applies despite the focusing sign. The ODE field satisfies $q_{tt} = q^3$ with the same data on $[-R, R]$.

Fix any $t_1 < \min\{T, T_*\}$. The assumed regularity makes the difference $w = u - q$ classical on the compact cone up to t_1. It satisfies

$$w_{tt} - w_{xx} = a(t, x)w, \qquad a = u^2 + uq + q^2.$$

The coefficient is bounded on that compact cone. Use the positive local energy

$$\mathcal{E}(t) = \frac{1}{2}\int_{-R+t}^{R-t} (w_t^2 + w_x^2 + w^2)\mathrm{d}x.$$

The moving-interval FTC gives

$$\begin{aligned}\mathcal{E}'(t) = &-\frac{1}{2}[(w_t - w_x)^2 + w^2](t, R - t) - \frac{1}{2}[(w_t + w_x)^2 + w^2](t, -R + t)\\ &+ \int_{-R+t}^{R-t} (a + 1)ww_t \mathrm{d}x \le C\mathcal{E}(t), \qquad C = \sup_{\text{cone}} |a + 1|.\end{aligned}$$

Here $2|ww_t| \le w^2 + w_t^2$ controls the last term. The initial energy is zero. Differentiating $e^{-Ct}\mathcal{E}(t)$ shows it is nonincreasing and nonnegative, hence zero. Thus $u = q$ on the cone up to every such t_1, in particular at its centre. Only a local positive comparison energy was used; q is not an L^2 field on the line.

For the second part, if $T_* > T$, continuity into H^1 would bound $\|u(t)\|_\infty$ on the compact interval $[0, T]$. But $u(t, 0) = \sqrt{2}/(T - t) \to \infty$ there, a contradiction. Hence $T_* \le T$ and

the local continuation alternative forces the full energy norm to diverge at its actual finite endpoint. If $T_* < T$, the comparison asserted equality only up to that common lifespan; it did not assert existence until T or locate the earlier blow-up. The cutoff produces genuine finite-energy line data, unlike a nonzero spatially constant ODE datum. The conservation part of Problem 1 also applies to this signed potential, but its positivity step fails: a conserved focusing energy cannot bound the full evolution norm.